



My Developer Journal

A Journal of Things I Learned on my journey as a developer

TECH

How to setup Tomcat Application on AWS EC2

Date: June 23, 2018 **Author:** Sunit Chatterjee ☐ 0 Comments

Amazon EC2 (Elastic Compute Cloud), is a service that provides scalable compute capacity in the cloud. It significantly reduces the time required to setup, configure and boot up a new server instances.

In this blog, we will demonstrate how to create a new EC2 instance and run a Tomcat Application Server on a EC2 Instance.

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Pre-requisites

1. We assume you already have an AWS account that you can use. Since this is only a demonstration blog, we will not use high performance servers and try to make maximum use of AWS Free Tier
2. You understand what a Tomcat Application server is

1.0 Creating an EC2 Instance

- Login to AWS Management Console – <https://console.aws.amazon.com/console/home>
- Go to >> Services >> Compute >> EC2 >> Create Instance >> Launch Instance

1.1 Choose an Amazon Machine Image (AMI)

- For our demo, we are selecting 'Amazon Linux' as an option



Amazon Linux 2 LTS Candidate 2 AMI (HVM), SSD Volume Type - ami-afd15ed0

Amazon Linux

Free tier eligible

Amazon Linux 2 LTS Candidate 2 provides an updated version of the Linux Kernel (4.14) tuned for E (gcc 7.3), an updated C runtime (glibc 2.26), newer tooling (binutils 2.29.1), and the latest software p

Root device type: ebs Virtualization type: hvm ENA Enabled: Yes

1.2 Choose Image Type

- Select size as `t2.micro`, which is the smallest size we can take for our demo.
- In actual deployments, please select a bigger size instance.

	General purpose	t2.micro Free tier eligible	1	1
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- **DO NOT** click on “**Review and Launch**”.
- Instead click on the button – “**Next: Configure Instance Details**”

1.3 Configure Instance Details

- In the next steps, configure the details for your instance and select following options
 - **Number of Instances** = 1
 - **Network** = Either select the already created VPC Or Create a New VPC
 - **Subnet** = Select any one of the available subnets. e.g – Subnet 1a | us-east-1a
 - **Assign Public IP**
 - Enable this option
 - We want to connect to our EC2 instance from outside of VPC (e.g from our machine using SSH, etc)
 - **Network Interface >> Primary IP** = Don't add anything (Auto assign IP)
 - Leave other options to their default values

Number of instances ⓘ [Launch into Auto Scaling Group](#) ⓘ

Purchasing option ⓘ ☐ Request Spot instances

Network ⓘ [Create new VPC](#)

Subnet ⓘ [Create new subnet](#)
59 IP Addresses available

Auto-assign Public IP ⓘ

Placement group ⓘ ☐ Add instance to placement group.

IAM role ⓘ [Create new IAM role](#)

Shutdown behavior ⓘ

Enable termination protection ⓘ ☒ Protect against accidental termination

Monitoring ⓘ ☐ Enable CloudWatch detailed monitoring
[Additional charges apply.](#)

Tenancy ⓘ [Additional charges will apply for dedicated tenancy.](#)

T2 Unlimited ⓘ ☐ Enable
[Additional charges may apply](#)

▼ Network interfaces ⓘ

Device	Network Interface	Subnet	Primary IP	Secondary IP
eth0	New network interface	subnet-8e1dc5a0	Auto-assign	Add IP

1.4 Configure Storage

- **Add Storage** = 10GB (which should be more than enough for our demo)
- **Select Volume Type** = General Purpose SSD

Volume Type ⓘ	Device ⓘ	Snapshot ⓘ	Size (GiB) ⓘ	Volume Type ⓘ
Root	/dev/xvda	snap-0d2c68a17b2738991	10	General Purpose SSD (GP2)

1.5 Add Tag

- Adding a tag helps identify the EC2 instance easily (when you create multiples of them)
- Add a Name Tag to the EC2 Instance

Step 5: Add Tags

A tag consists of a case-sensitive key-value pair. For example, you could define a tag with key = Name and value = Webserver.

A copy of a tag can be applied to volumes, instances or both.

Tags will be applied to all instances and volumes. [Learn more](#) about tagging your Amazon EC2 resources.

Key (127 characters maximum)	Value (255 characters maximum)	Instances 	Volumes 	
Name	Application Server 1	☒	☒	

1.6 Security groups

- This is a very important security step.
- It controls the security of the EC2 instances and also defines who can access it and how.
- Create a new Security Group, and allow access on following ports
 - **SSH Access**
 - This type of access is required to connect to the EC2 instance and run commands / install softwares, etc
 - Port = 22
 - Allow access from “My IP”.
 - It will allow access from you current machine IP Address
 - **TCP Access**
 - Since we are planning to install Tomcat Application server which runs on Port 8080, we will allow access to this port
 - This is only for demonstration and testing purpose
 - Ideally in a real use case, we will not expose the application server port directly, but instead it will be accessed through a Load Balancer or a Web Server.
 - Port = 8080
 - Allow from anywhere

Assign a security group: ☒ Create a **new** security group
☐ Select an **existing** security group

Security group name:

Description:

Type i	Protocol i	Port Range i	Source i	Description i
SSH ▾	TCP	22	My IP ▾ 203.90.98.60/32	SSH Access
Custom TCP F ▾	TCP	8080	Anywhere ▾ 0.0.0.0/0, ::/0	Temporary Access

Add Rule

1.7 Creating Key Pair

- In order to SSH to the EC2 instance, you will need a **private key**.
- To generate and download the Private Key
 - Create a new Key Pair (e.g with name as “AppServerKey”)
 - Download Key Pair as **AppServerKey.pem**

Select an existing key pair or create a new key pair ×

A key pair consists of a **public key** that AWS stores, and a **private key file** that you store. Together, they allow you to connect to your instance securely. For Windows AMIs, the private key file is required to obtain the password used to log into your instance. For Linux AMIs, the private key file allows you to securely SSH into your instance.

Note: The selected key pair will be added to the set of keys authorized for this instance. Learn more about [removing existing key pairs from a public AMI](#).

Create a new key pair ▾

Key pair name

Download Key Pair

 You have to download the **private key file** (*.pem file) before you can continue. **Store it in a secure and accessible location.** You will not be able to download the file again after it's created.

- Save this file properly, as you won't be able to generate it again.

2.0 Connecting to EC2 Instance

To run commands within the EC2 Instance, you will need to connect to it. If you are on Linux box, you can directly run following commands to establish a SSH connection to the EC2 Instance

- Change permission of the Private Key File (created in previous step)

```
sudo chmod 600 AppServerKey.pem
```

- Find out the Public DNS of the EC2 Instance
e.g *ec2-54-92-184-62.compute-1.amazonaws.com*
- Default user created is **ec2-user**
- You can now connect using the following SSH command

```
ssh -i @
```

e.g

- `ssh -i "AppServerKey.pem" ec2-user@ec2-54-92-184-62.compute-1.amazonaws.com`

If you are on Windows, then you can use PuTTY, to connect to EC2 Instance

3.0 Install Java

There are two options for installing JDK

1. Install Open-JDK
2. Install Oracle JDK

We will go with second option and Install JDK 1.8. (You can install any other version too)

Connect to your EC2 instance by SSH and run following steps.

3.1 Create Folder for download

Create a folder for Downloads (option step)

```
mkdir -p Downloads  
cd Downloads/
```

3.2 Download Oracle JDK

You can download using the command below

```
wget --no-check-certificate -c --header "Cookie:  
oraclelicense=accept-securebackup-cookie"
```

E.g to download JDK 1.8 we can run following

```
wget --no-check-certificate -c --header "Cookie:  
oraclelicense=accept-securebackup-cookie"  
http://download.oracle.com/otn-pub/java/jdk/8u171-  
b11/512cd62ec5174c3487ac17c61aaa89e8/jdk-8u171-linux-  
x64.tar.gz
```

3.3 Installing Oracle JDK

Run below commands to Install Java

```
cd /home/ec2-user/Downloads
sudo su

#Create the folder
mkdir /opt/java

#Copy and Extract JDK
cd /opt/java
cp /home/ec2-user/Downloads/jdk-8u171-linux-x64.tar.gz .
tar -xvf jdk-8u171-linux-x64.tar.gz
rm -f jdk-8u171-linux-x64.tar.gz

#update alternatives for java/javac/jar commands
update-alternatives --install /usr/bin/java java
/opt/java/jdk1.8.0_171/bin/java 100
update-alternatives --install /usr/bin/javac javac
/opt/java/jdk1.8.0_171/bin/javac 100
update-alternatives --install /usr/bin/jar jar
/opt/java/jdk1.8.0_171/bin/jar 100

#Add environment variables
cd /etc/profile.d/
touch jdk.sh
echo "export JAVA_HOME=/opt/java/jdk1.8.0_171/" >> jdk.sh
echo "export JRE_HOME=/opt/java/jdk1.8.0_171/jre" >> jdk.sh
echo "export
PATH=$PATH:/opt/java/jdk1.8.0_171/bin:/opt/java/jdk1.8.0_171/j
re/bin" >> jdk.sh
```

Once above steps are done, reboot your EC2 Instance.

4.0 Download and Install Tomcat

Connect to your EC2 instance by SSH and run following steps.

4.1 Create a new user for Tomcat

- Create a new user – *tomcat*
- This is an optional step, and **you may also choose to skip it.**

```
sudo su
adduser tomcat
passwd tomcat
```

4.2 Download and setup Tomcat

For our demo purpose, we will use Tomcat 9.0.8

Download and extract Tomcat 9.0.8 to the home folder of the user

```
cd /home/ec2-user/Downloads

#Download Tomcat 9.0.8
wget http://www-eu.apache.org/dist/tomcat/tomcat-
9/v9.0.8/bin/apache-tomcat-9.0.8.tar.gz

#Extract Tomcat
tar xvf apache-tomcat-9.0.8.tar.gz

#Move the extracted tomcat folder to home directory of the
user
mv apache-tomcat-9.0.8 /home/ec2-user

#Add environment variables for CATALINA_HOME
cd /etc/profile.d/
touch tomcat.sh
echo "export CATALINA_HOME=/home/ec2-user/apache-tomcat-
9.0.8".sh
```

Now the setup of tomcat is complete, and it can be run now.

5.0 Add startup Script for Tomcat Application

There is one more thing we need to do.

Generally we can start our tomcat server manually by running the `startup.sh` located in `CATALINA_HOME/bin` folder

However it will be too cumbersome to manually start Tomcat when our EC2 server is started. Ideally it would be good if Tomcat application auto-starts whenever our EC2 instance is started.

For this we will create a Tomcat Service and add it to the Linux Startup

- Create a file with name *tomcat.service*, within `/etc/systemd/system`

```
sudo nano /etc/systemd/system/tomcat.service
```

- Add below script to the above file – `tomcat.service` .
Make sure to set the Tomcat and JDK path correctly.

```
# Reference : http://techkube.com/article/how-install-
tomcat-ec2-instance
[Unit]
Description=Apache Tomcat
After=network.target

[Service]
Type=forking

Environment=CATALINA_PID=/home/ec2-user/apache-tomcat-
9.0.8/temp/tomcat9.pid
Environment=JAVA_HOME=/opt/java/jdk1.8.0_171/
Environment=CATALINA_HOME=/home/ec2-user/apache-tomcat-
9.0.8
Environment=CATALINA_BASE=/home/ec2-user/apache-tomcat-
9.0.8
Environment="JAVA_OPTS=-Dfile.encoding=UTF-8 -
Djava.awt.headless=true"

ExecStart=/home/ec2-user/apache-tomcat-9.0.8/bin/startup.sh
ExecStop=/home/ec2-user/apache-tomcat-9.0.8/bin/shutdown.sh

User=ec2-user
Group=ec2-user

[Install]
WantedBy=multi-user.target
```

- Run below commands to run Tomcat as a service whenever the current EC2 instance starts.

```
sudo systemctl daemon-reload
sudo systemctl start tomcat
sudo systemctl enable tomcat
```

- You can manually start and stop tomcat using below commands also

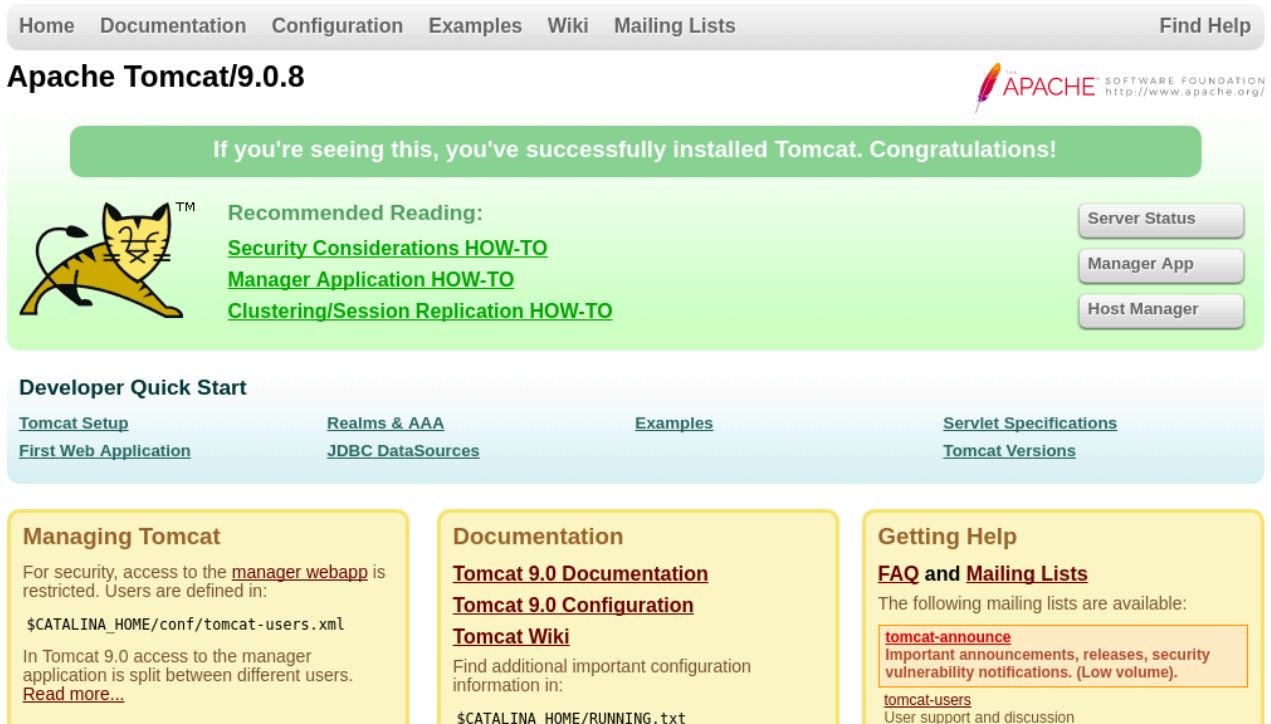
```
sudo service tomcat start
sudo service tomcat stop
```

6.0 Testing Tomcat Application

Now start the tomcat application, and check if its running

Connect to `http://:8080` to access the Tomcat Root application

e.g. On accessing `http://ec2-54-234-212-57.compute-1.amazonaws.com:8080/`, we should see a similar page as below



The screenshot shows the Apache Tomcat 9.0.8 root page. At the top, there is a navigation bar with links: Home, Documentation, Configuration, Examples, Wiki, Mailing Lists, and a Find Help button. Below the navigation bar, the title "Apache Tomcat/9.0.8" is displayed. To the right of the title is the Apache Software Foundation logo and the URL <http://www.apache.org/>. A green banner in the center reads: "If you're seeing this, you've successfully installed Tomcat. Congratulations!". Below the banner, on the left, is the Tomcat logo (a yellow cat). To the right of the logo, under the heading "Recommended Reading:", are three links: [Security Considerations HOW-TO](#), [Manager Application HOW-TO](#), and [Clustering/Session Replication HOW-TO](#). To the right of these links are three buttons: "Server Status", "Manager App", and "Host Manager". Below the banner, there is a "Developer Quick Start" section with four links: [Tomcat Setup](#), [Realms & AAA](#), [Examples](#), and [Servlet Specifications](#). Below these links are four sub-sections: "Managing Tomcat", "Documentation", "Getting Help", and "FAQ and Mailing Lists". Each sub-section contains text and links to further resources.

This completes our blog on setting up Tomcat Application server on AWS EC2.



Published by Sunit Chatterjee

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